Comp. Phys. Lab for Quantum Mechanics
Winter Quarter 2026
Physics 115L

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Physics 317

Lecture: Tuesday 2:10-3:00 PM in Roessler 55

-**Textbooks:** Intro. to Quantum Mechanics (3rd Ed.) by Griffiths and Schroeder
- (Recommended for reference, or any other Intro. QM text.)
Lecture notes on Scientific Python (link on course site)

Course TA: TBD tbd@ucdavis.edu

Office Hours: Mulhearn: Mondays 11:00 AM-12:00 PM

Exams/Quizzes: None planned, but see caveats in the plagiarism section.
Final Project: Due December, 10 2025 10:30 AM

Course Description:
Applications of computational physics to problems from Quantum Mechanics. Application of numerical techniques, and comparison to analytic solutions (where they exist).

System of Units:
To receive any credit, your solutions must adopt the system of units as presented in lecture and in the write-up for each assignment. The system of units must be used throughout your solution. You may not e.g. do the calcuation in SI and then convert to our system of units later. If you see physical constants like Planck's constant appearing your code, that is likely a misunderstanding!

Plagiarism and Academic Conduct:
You must adhere to the UC Davis Code of Academic Conduct, available online at ossja website. In particular, you may not take credit for any work not created by you. **You may not direct search engines, generative AI tools, homework help websites, or other external resources toward the specific problems that are assigned to you.** If you do, then you must disclose it, and you will receive a zero for the entire assignment. If you do not disclose it, and are caught, then you will fail the entire course.

You are allowed to collaborate with other students in the course, including discussing specific problems in the homework, unless you have reason to believe the students are using unauthorized methods. The solution that you provide must be your own: each and every line. In general, you should be able to explain why each line in your code is there. If you do not understand something in your solution because you relied on trial and error, then you should state that in the comments, including explaining any other things that you tried. For example:

```
# I don't understand why this factor of two is needed
# I have included it because it gives the correct result:
```

If I become suspicous that plagarism or cheating is occurring, I reserve the right to add quizes, exams, or other means of grading. You should always be prepared to explain your code if asked.

Course Philosophy:
Computational physics is essential to nearly every career related to physics. Unfortunately, the educational pipeline, since grade school, does not adequately prepare students with programming skills, and so a central feature of computational physics courses is a wide distribution of student programming abilities. There are students that are better programmers than most physics professors, students that have little practice outside of the required computational courses, and everything in between.

Fortunately, in this course we are concerned with **computational physics** and not just **computer programming**. You do not need to be an expert computer programmer to succeed in this course. You do need to gain enough programming competency to handle the fascinating physics problems we will tackle. **The primary objective of this course is to make you confident in using programming independently as an essential and effective tool for solving physics problems.**

This is a one unit course, and I will do my best to keep the workload appropriate. However, given the wide variation in student programming abilities, inevitably some students will find the homework more time-consuming than others. If you do find yourself working longer on the assignments, please understand that this is time well-spent: gaining more proficiency in computing is almost certainly one of the best possible uses of your time at this stage in your career. There are a vanishing small number of physics-related career options that do not require programming ability.

Homework:
There will be up four home-work assignments which will feature problems in computational physics.

Final Project:
In lieu of a final exam, you will design an independent creative problem. While your particular problem may have already been solved online, your solution should vary substantially from anything available online.

Grades:
Final grades will be approximately 80% Homework and 20% Final Project. You will be allowed to submit one homework assignment late, but you must complete every assignment and your final project to complete the course.

Contacting the Instructor:
I have posted my email, but that is mainly so that you can contact me after the course has ended. During the course, the most reliable way to reach me is by talking to me after lecture or coming to office hours. If you must reach me remotely, please use the course website instead of email.